Midterm 2 — Sample 1

DATA 412/612

Practice exam, questions only. Weeks 1–11. The material since Midterm 1 is joins, strings, factors, dates, and statistics (a two-sample test and a linear model).

Put your answers in an R Markdown document and keep the code visible (`knitr::opts_chunk$set(echo = TRUE)` in the setup chunk). You may use your notes and the course website. You may not use Google, any other website, or an LLM. Do not discuss the exam with anyone else. Load the tidyverse. `mpg` comes with `ggplot2`.

1. **Factors** (8 pts). In `mpg`, lump every `class` that appears fewer than 20 times into `"Other"`. Reorder the levels of that factor by the median `hwy`, lowest median first. Print the level order and a count of each level.

2. **Dates** (8 pts). Parse `stamp` as a date-time in the time zone `America/New_York`. Add the weekday name, the hour in that time zone, and the same instant displayed in `UTC`. Then add two days with a period, not a duration, and show the new calendar date.

```
stamps <- tibble(
  stamp = c("2024-03-10 01:30", "2024-07-04 18:05")
)
```

3. **A two-sample test** (8 pts). Using `mpg`, compare highway mileage for front-wheel drive (`drv == "f"`) with rear-wheel drive (`drv == "r"`). State the null hypothesis in one sentence. Report the difference in means and the p-value. Do not drop cars by hand before the test; let the test see those two groups.

4. **A linear model** (10 pts). Fit `hwy` on `displ` in `mpg`. In one sentence, say what the `displ` coefficient means. Plot residuals against fitted values and say whether the cloud looks like random scatter around zero.

5. **A join** (6 pts). Load `nycflights13`. Add the airline name onto `flights` and report the mean arrival delay for each airline name, ignoring missing delays. Show the three names with the largest mean delay.